

③

a)
$$W_{a \rightarrow b} = \Delta K$$
$$= 1,5 \cdot 10^{-6} \text{ [J]}$$

b)
$$q(V_a - V_b) = W_{a \rightarrow b}$$

$$V_a - V_b = \frac{1,5 \cdot 10^{-6}}{4,2 \cdot 10^{-9}}$$

$$= 357,142 \text{ [V]}$$

c)
$$W_{a \rightarrow b} = \int_a^b q \vec{E} \cdot d\vec{\ell}$$

$$\frac{W_{a \rightarrow b}}{q} = \int_a^b E d\ell \cos 0$$

$$V_a - V_b = E d_{ab}$$

$\Rightarrow$

$$E = \frac{357,142}{0,06}$$

$$= 5,952 \cdot 10^3 \text{ [N/e]}$$

